ACCIVIS

An Accident Visualization System

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VISUAL ANALYTICS COURSE

Goal of the visualization

The main purpose is to find potential causes of accidents, in order to prevent them

So the purpose of the visual analytic system is to:

- Provide an overview of the accidents
- Explore details of accidents with the aid of interaction

The intended users of the system are members of the Ministry of Infrastructure (ci frega effettivamente dirlo?)

Structure of data

The dataset of accidents occurred in Italy in 2024 was taken from ISTAT.

Attribute	Description
Area	ID of Region
Road section	Number identifying a road section. Identifies: tunnel, bend, crossroad, traffic circle, bump/slope or bottleneck, straight stretch, level crossing.
Accident type	Number identifying the accident type. Identifies: accidents between vehicles, vehicle-pedestrian accidents and accidents involving a single vehicle.
Deadliness	Boolean attribute that identifies if the accident involved death or only injury
Hour of accident	Time of accident measured with hour resolution
Month	Month number
Weekday	Number identifying the weekdays: monday, tuesday, wednesday, etc...
Observations	Number of accidents occurred with the specified attributes

The dataset used for the visualization presents 92.402 tuples organized in 8 attributes.

Chosen visualizations

The system is logically divided into four areas:

Geographical distribution:

- Choropleth map
- Bar chart (redundant chart)

Temporal distribution:

- Heatmaps

Categorical distribution:

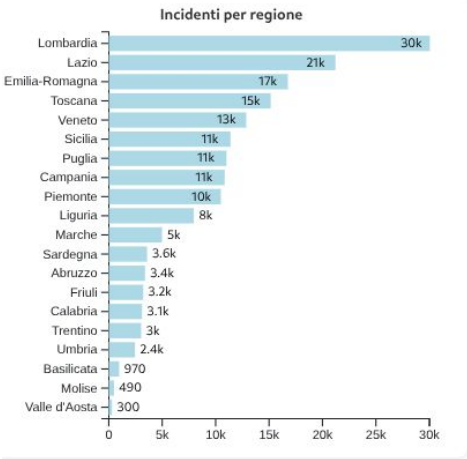
- Bar charts

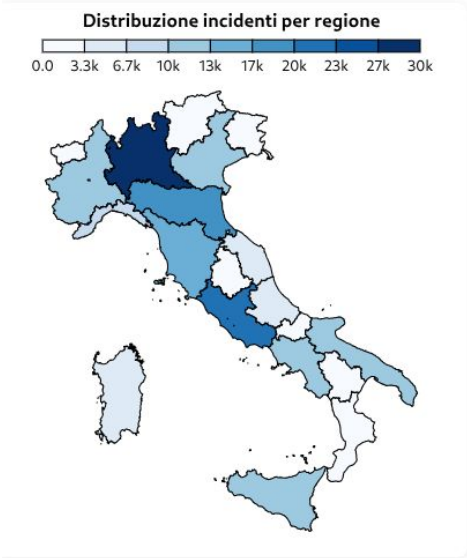
PCA

- Scatterplot with color encoding

Geographical distribution

- Checking distribution of accidents over regions
- Redundant chart for keeping an overview and making comparisons easier

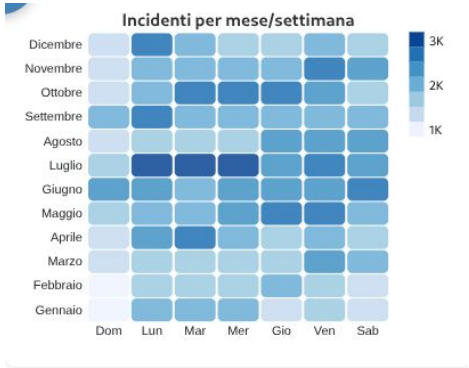


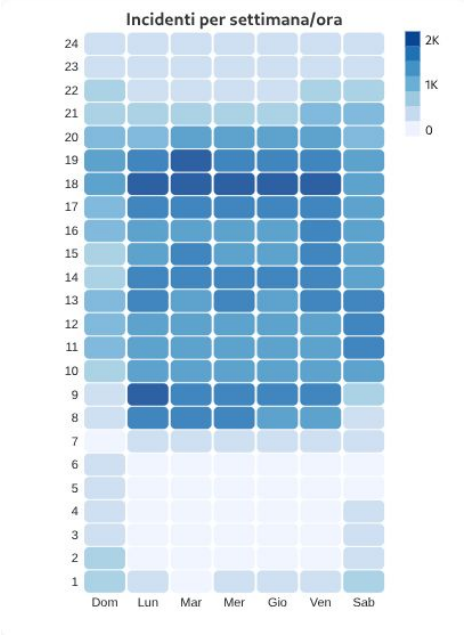


Temporal distribution

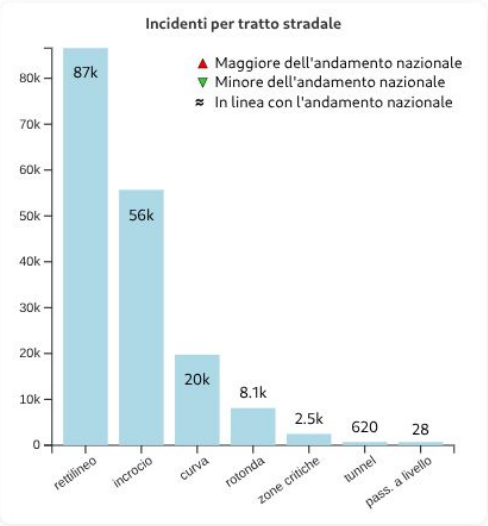
- Checking distribution of accidents over hours/weekdays

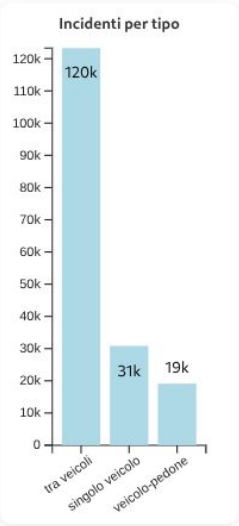
- Finding temporal patterns

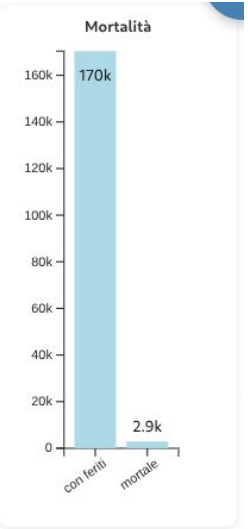




Categorical distribution

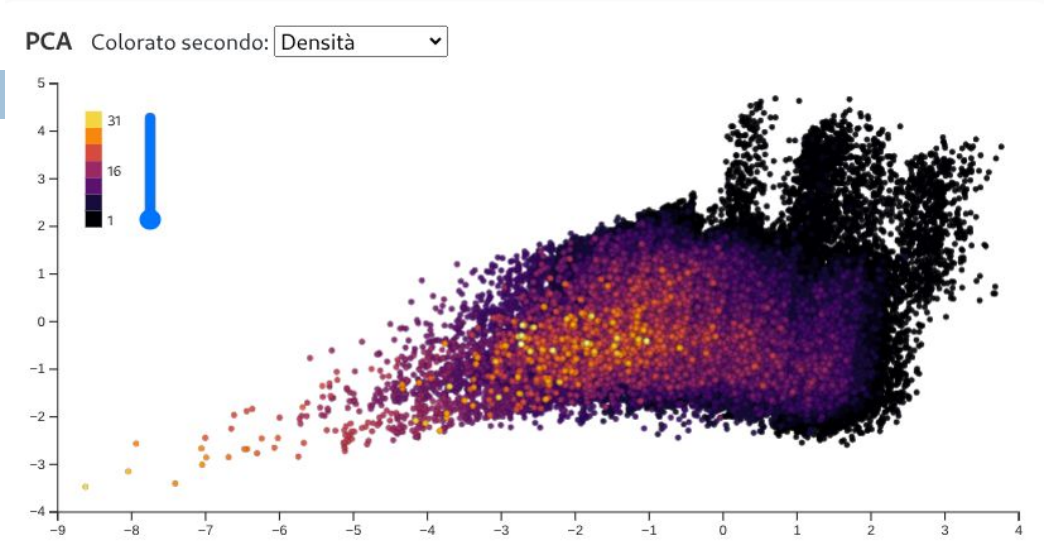






- Sorted bars
- Double encoding for more accurate comparisons

PCA



- Checking clusters of similar accidents*
- Density map
- Coloring according to attributes

DEMO